Notes: Engelberg and Parsons (JF, 2011)

The Causal Impact of Media in Financial Markets

Contents

1 Big picture

- **Question.** Does media coverage itself causally affect investor behavior, or does media merely move in parallel with the underlying events that already determine trading?
- **Setting.** The paper studies earnings announcements of S&P 500 firms and compares how retail investors in 19 U.S. cities react when *their own local newspaper* does or does not cover the announcement.
- **Core identification problem.** Media coverage is endogenous. Newspapers choose which earnings announcements to cover, and that choice may reflect the same latent features of an event that also make local investors want to trade.
- **Main idea.** Hold the underlying information event fixed and vary media exposure across cities. For the same earnings release, Houston investors may see a story in the *Houston Chronicle* while San Antonio investors may not see a story in the *Express-News*. Cross-city differences in trading around the same event then identify media effects.
- **Main empirical strategy.** Combine (i) S&P 500 earnings announcement dates, (ii) newspaper stories from ProQuest, and (iii) account-level retail brokerage data with zip codes. Run city–firm–date panel regressions of local trading on local newspaper coverage, then test credibility with local-demand controls, pairwise fixed effects, extreme-weather transmission shocks, and story-timing falsification tests.
- **Main quantitative result.** Local newspaper coverage raises local retail trading substantially. Depending on the specification, the effect ranges from about 8% in the most saturated fixed-effect design to nearly 50% in the main controlled specification.
- **Main mechanism result.** The effect is not confined to buys. It also appears for sells, though somewhat more strongly for buys. So the media is not simply producing a pure "buying attention" effect; it raises trading activity more broadly.
- **Most convincing causal evidence.** When bad local weather likely disrupts physical newspaper delivery, the local-coverage effect collapses. And in day-by-day regressions, only coverage on the *same day* predicts trading on that day; future coverage does not predict current trading.
- **Why this matters.** This is one of the cleanest "same facts, different exposure" papers in empirical finance. It shows that media can affect investor behavior even after one holds the underlying information event fixed.
- **Economic takeaway.** Media is not just a passive mirror of financial news. It is part of the transmission technology through which information reaches investors and becomes market activity.

2 Data and measurement (key objects)

2.1 Three core data sources

- **Earnings announcements.** All S&P 500 earnings announcements from January 1991 to December 2007. Announcement dates are taken from COMPUSTAT and confirmed with I/B/E/S.

- **Media data.** ProQuest archives for 19 local newspapers, plus *USA Today* and the *Wall Street Journal*.
- **Retail trading data.** The large discount brokerage data from Barber and Odean. Daily trades are available from January 1991 to November 1996, while holdings are available monthly from January 1991 to December 1996.

2.2 Trading markets and local newspapers

- The paper defines 19 mutually exclusive local trading regions around major U.S. cities, using investors' zip codes.
- Each city is paired with a local daily newspaper. The set includes papers such as the *Boston Globe*, *Detroit News*, *Houston Chronicle*, *Los Angeles Times*, *Minneapolis Star Tribune*, *New York Times*, *San Francisco Chronicle*, *Seattle Post Intelligencer*, *Washington Post*, and others.
- The key identifying assumption is modest: investors in a city are *more likely* to read their own local paper than a remote city's paper. The design does *not* require local newspapers to be the investors' only source of financial information.

2.3 Units of observation and main variables

- **Unit of observation.** City $\times$ firm $\times$ earnings-announcement date.
- **Local Media Coverage.** Dummy equal to one if the newspaper associated with city i writes a story about firm j on day 0, 1, or 2 after the earnings announcement.
- **Baseline dependent variable.** $\log(1 + \text{Local Dollar Trading Volume})$, where trading is aggregated across all local investors in the city over the 3-day window after the earnings release.
- **Alternative dependent variables.**
  - $\log(1 + \text{Number of Accounts Traded})$,
  - a dummy for whether there is *any* local trading at all.
- **Controls.** Firm size, earnings surprise quintiles, newspaper fixed effects, city fixed effects, date fixed effects, and in richer specifications additional measures of local demand and pairwise fixed effects.

2.4 Useful descriptive facts

- The brokerage sample begins with 77,795 households. Of these, 54,297 have valid zip codes, 43,198 hold at least one common stock that can be matched to CRSP/COMPUSTAT, and 15,951 are within 100 kilometers of one of the 19 newspaper headquarters.
- The average household holds about 2.26 stocks and trades about 1.75 stocks per month.
- The mean of **Local Media Coverage** is only 0.028, so roughly 1 in 35 quarterly earnings announcements appears in a local paper.
- By contrast, the *Wall Street Journal* reports an earnings story about 29% of the time in the trading sample and about 33% over the broader 1991–2007 coverage period.

- On days when earnings are covered by the local paper, average local absolute dollar trading is about \$2,200; on local non-news days it is only about \$290.

2.5 Local bias in holdings and local bias in coverage

- Investors show strong home bias. Minneapolis-area accounts hold about 21.8% of local stocks versus 3.6% of nonlocal stocks; San Francisco-area accounts hold about 44.4% of local stocks versus 15.1% of nonlocal stocks.
- Newspapers are also strongly local. The *San Antonio Express-News*, for example, covers only about 3.5% of nonlocal earnings announcements but more than 40% of local ones.
- The *New York Times* is the least locally tilted paper in the sample, yet even it covers New York firms about twice as often as remote firms.

Interpretation. These local tilts are exactly what make the design possible. But they are also what make identification nontrivial, because local interest in a stock could drive both coverage and trading.

2.6 What is observed and what is not

- **Observed directly.** Earnings dates, newspaper stories, local trading volume, local holdings, lagged local trading, market capitalization, earnings surprises, and local weather.
- **Not observed directly.** Investors' true information sets, exact readership, newspapers' internal editorial rules, and all latent city–firm interests that might influence both coverage and trading.
- **Why this matters.** The paper's entire identification strategy is about purging or bypassing these latent confounds.

3 Models and empirical specifications (all equations + interpretation)

3.1 The identification problem in conceptual form

The paper writes investor demand as

$$D(X, M(X, Y)), \tag{1}$$

where X affects both media coverage and investor demand, while Y affects only media coverage. Taking a total derivative gives

$$dD = \frac{\partial D}{\partial X}dX + \frac{\partial D}{\partial M}\frac{\partial M}{\partial X}dX + \frac{\partial D}{\partial M}\frac{\partial M}{\partial Y}dY. \tag{2}$$

Interpretation.

- The first term is the direct effect of the underlying event.

- The second term captures the fact that media selection may respond to the same event characteristics that also move investors.
- The third term is the "pure" media-transmission term driven by media-specific variation, such as delivery or timing.

The empirical goal is to identify the latter two media-related terms separately from the direct effect of fundamentals.

3.2 Baseline regression

The main specification is

$$\begin{aligned}\log\bigl(1+\text{Local Trading Volume}_{i,j,t}\bigr) = \alpha &+ \delta\,\text{Local Media Coverage}_{i,j,t} \\ &+ \phi\,\text{Firm Attributes}_{j,t} + \zeta\,\text{Earnings Surprise}_{j,t} \\ &+ \gamma\,\text{Media Fixed Effects}_{i} + \sigma\,\text{City Fixed Effects}_{i} + \varepsilon_{i,j,t}. \quad (1)\end{aligned}$$

The dependent variable is measured over a 3-day window after the earnings release.

Interpretation.

- Because the event is an earnings release for a given firm on a given date, the comparison is across local markets exposed to different local-paper coverage of the *same* event.
- Newspaper fixed effects absorb the average influence of each paper, while city fixed effects absorb average differences in trading activity across markets.
- The coefficient on `Local Media Coverage` is therefore interpreted as the extra effect of that paper being *local* to the investors in that city.

3.3 Controls and why they matter

- **Firm size.** Larger firms generate more trading.
- **SUE quintiles.** The paper controls for standardized unexpected earnings so that more surprising announcements can generate more trading even absent media.
- **Paper dummies.** Local and national-paper coverage dummies allow the authors to distinguish the incremental local effect from any general informational effect of being covered at all.
- **Industry, city, and date fixed effects.** These soak up broad heterogeneity in firms, cities, and aggregate time-series trading conditions.

An important empirical fact from the paper is that the media effect is much larger than the direct information effect captured by SUE. In the reported specifications, the largest SUE coefficient never exceeds about 0.15, while the local-media coefficient remains 3 to 10 times larger in magnitude.

3.4 Buy and sell decompositions

The paper re-runs the main regression separately for buy-side and sell-side local volume.

Interpretation.

- If media mainly works through attention to positive stories, one might expect only buys to react.
- Instead, the paper finds clear effects on both buys and sells. That is more consistent with local media increasing processing, salience, and willingness to act on the information rather than simply creating net bullishness.

3.5 Augmenting with preexisting local demand

To address the concern that local papers simply cover the firms that local investors already care about, the paper adds proxies for local demand:

$$\begin{aligned}\log(1+\text{Trade}_{i,j,t}) = \cdots &+ \theta_1\,\text{Local Firm Dummy}_{i,j}\\ &+ \theta_2\,\text{Fraction of Accounts Holding}_{i,j,t-1} + \theta_3\,\text{Fraction of Accounts Trading}_{i,j,t-1} + \varepsilon_{i,j,t}.\end{aligned} \tag{3}$$

Interpretation.

- `Local Firm Dummy` captures simple geography and home bias.
- `Fraction of Accounts Holding` captures the local investor base already exposed to the stock.
- `Fraction of Accounts Trading` captures very recent local attention and habit in that stock.

If local coverage were merely reflecting preexisting local demand, these controls should wipe out the media coefficient. They do not.

3.6 Weather as exogenous variation in media transmission

The weather specification can be summarized as

$$\begin{aligned}\log(1+\text{Trade}_{i,j,t}) = \cdots &+ \delta\,\text{Local Media Coverage}_{i,j,t} + \kappa\,\text{Extreme Weather}_{i,t}\\ &+ \eta\,(\text{Extreme Weather}_{i,t} \times \text{Local Media Coverage}_{i,j,t}) + \varepsilon_{i,j,t}.\end{aligned} \tag{4}$$

The key weather variables are local hail, local snow/blizzards, and an aggregate extreme-weather dummy.

Interpretation.

- Weather alone should not matter much for local trading in these regressions.
- What matters is the interaction: if a local paper *did* report the story but delivery was disrupted, then investors should not respond to the content.
- A large negative interaction is therefore strong evidence that the paper reaches investors through the local newspaper channel, not through some omitted city–firm variable.

3.7 Pairwise fixed effects

A sharper identification design adds pairwise fixed effects:

$$\log(1+\text{Trade}_{i,j,t}) = \delta\,\text{Local Coverage}_{i,j,t} + \mu_{ij} + \nu_{jt} + \omega_{it} + \varepsilon_{i,j,t}. \tag{5}$$

where μ_{ij} are firm–city fixed effects, ν_{jt} are firm–date fixed effects, and ω_{it} are city–date fixed effects.

Interpretation.

- **Firm–city fixed effects** absorb persistent local bias toward a firm.
- **Firm–date fixed effects** absorb any firm-specific national news that moves all cities on that date.
- **City–date fixed effects** absorb local shocks to trading activity on a particular day.

This is an extremely demanding specification because the coefficient is identified only from residual variation in whether a given city's local paper covered the same firm on the same date, relative to other cities and after sweeping out all pairwise patterns.

3.8 Timing-to-print falsification

The final specification uses day-by-day timing variation:

$$\begin{aligned}\log(1 + \text{Trade}_{i,j,t+\tau}) = {} & \delta_0 \, \text{Coverage on Day } 0_{i,j,t} + \delta_1 \, \text{Coverage on Day } 1_{i,j,t} \\ & + \delta_2 \, \text{Coverage on Day } 2_{i,j,t} + \mu_{ij} + \nu_{jt} + \omega_{it} + \varepsilon_{i,j,t+\tau}. \end{aligned} \tag{6}$$

for $\tau \in \{0, 1, 2\}$.

Interpretation.

- Same-day coverage should predict same-day trading if newspapers causally move investor behavior.
- Future coverage should *not* predict current trading. If it did, that would signal misspecification or omitted demand shocks.
- This timing test is especially persuasive because it uses very high-frequency variation in when the exact same story reaches different cities.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no classical instrument here

- The paper is not an IV design in the narrow sense.
- Its identification comes from a **same information event, different media exposure** design.
- The strength of the paper comes from piling multiple sources of credibility on top of this design rather than relying on one exclusion restriction.

4.2 Why the cross-sectional comparison is powerful

- Standard event studies compare market outcomes before and after a news event, which cannot separate the underlying event from the act of media transmission.

- This paper instead compares *different local investor groups* reacting to the same earnings release.
- That is conceptually similar to designs in media economics where the same national content reaches different audiences with different intensity.

4.3 What exactly could still go wrong?

The main alternative story is:

> Local newspapers cover exactly those firms that local investors were already unusually interested in, and local investors trade for that reason rather than because of the newspaper.

This concern is serious because local investors have home bias and local papers clearly favor local firms.

4.4 Why the paper's answers are convincing

- **Answer 1: direct controls for local demand.** Once the paper controls for local-firm status, local holdings, and lagged local trading, the media coefficient falls but remains large and highly significant.
- **Answer 2: fixed effects.** The firm–city, firm–date, and city–date fixed effects leave very little scope for omitted persistent local preferences or common firm news.
- **Answer 3: transmission shocks.** Extreme weather changes whether the newspaper actually reaches readers, but leaves the content and basic local demand unchanged.
- **Answer 4: timing falsification.** Only the exact reporting day predicts the exact trading day. Future stories do not predict earlier trading.

4.5 What makes the estimates especially credible

- The sample period is pre-Internet, which is useful because physical newspaper delivery was more central and easier to disrupt.
- The paper does not rely on one single reduced-form coefficient. It documents the same qualitative effect in baseline regressions, buy/sell splits, local-demand controls, weather interactions, alternative dependent variables, saturated fixed effects, and timing tests.
- The interpretation is disciplined: the authors do *not* claim that all investors read only local papers, only that local exposure is higher locally than nonlocally.

4.6 Main limitations

- The analysis is about **retail trading volume**, not price efficiency or long-run welfare.
- The investors come from one discount brokerage sample, so external validity to all households or institutions is limited.

- The mechanism is inferred rather than observed: the paper shows that media matters, but not whether it works through attention, persuasion, interpretation, salience, or reduced information-processing costs.
- Because the data are mostly pre-Internet and pre-social-media, the quantitative magnitude may not map directly to the current media environment.

4.7 Relation to the literature

- Relative to papers like Huberman and Regev, this paper moves from anecdotal examples to a systematic multi-event identification design.
- Relative to investor-attention papers, its distinctive contribution is causality: the same underlying event reaches different local investors through different newspaper channels.
- Relative to media-economics papers in political economy, it brings the "different exposure to the same content" idea into financial markets.

5 Tables

5.1 Table I: summary statistics and the local-bias fact

Table I

Summary Statistics

Account and trading data are taken from the large discount brokerage database with demographic information in Barber and Odean (2000). Accounts in Area is the number of accounts within 100 kilometers of the city's local paper headquarters. Local stocks are stocks of firms within 100 kilometers of the city's local paper headquarters. A paper "covers" an earnings announcement in our sample if a story about that firm appears on day 0, 1, or 2 after the firm's earnings announcement in the ProQuest database. Earnings announcement dates are taken from COMPUSTAT and confirmed with I/B/E/S. Pr(covering a nonlocal EA) is the frequency in which that paper covered a nonlocal earnings announcement. Pr(covering a local EA) is the frequency in which that paper covered a local earnings announcement.

	Accounts				Paper	
	Accounts in Area	Avg # of Stocks Held	Local Stocks Held / Local Stocks Avail	Nonlocal Stocks Held/ Nonlocal Stocks Avail	Pr(Covering a Local EA)	Pr(Covering a Nonlocal EA)
Atlanta (*Journal Constitution*)	398	2.35	0.135	0.035	0.716	0.019
Boston (*Globe*)	635	2.12	0.137	0.050	0.225	0.015
Denver (*Post*)	488	2.08	0.109	0.041	0.263	0.015
Detroit (*News*)	317	2.63	0.130	0.036	0.317	0.006
Houston (*Chronicle*)	607	2.18	0.164	0.047	0.372	0.011
Las Vegas (*Review Journal*)	167	2.20	0.087	0.018	0.282	0.010
Los Angeles (*Times*)	1,913	2.13	0.218	0.112	0.279	0.047
Minneapolis (*Star Tribune*)	445	2.08	0.218	0.036	0.305	0.012
New Orleans (*Times Picayune*)	134	2.41	0.086	0.016	0.657	0.006
New York (*Times*)	2,808	2.28	0.211	0.152	0.300	0.154
Pittsburgh (*Post Gazette*)	318	2.28	0.151	0.031	0.365	0.038
Sacramento (*Bee*)	500	2.20	0.128	0.077	0.011	0.002
San Antonio (*Express News*)	142	2.31	0.091	0.016	0.401	0.035
San Diego (*Union Tribune*)	517	2.14	0.161	0.047	0.500	0.029
San Francisco (*Chronicle*)	4,076	2.17	0.444	0.151	0.260	0.008
Seattle (*Post Intelligencer*)	1,019	2.11	0.301	0.060	0.461	0.013
St. Louis (*Post Dispatch*)	241	2.06	0.143	0.021	0.628	0.008
St. Petersburg (*Times*)	243	2.52	0.069	0.028	0.243	0.015
Washington D.C. (*Post*)	983	2.42	0.188	0.079	0.513	0.029
USA *Today*	–	–	–	–	0.023	0.041
Wall Street Journal	–	–	–	–	0.438	0.318

(*continued*)

Table I—*Continued*

	Mean	Standard Deviation	1st Percentile	10th Percentile	Median	90th Percentile	99th Percentile
Market capitalization (in millions)	5,660	10,676	62	466	2,513	12,825	54,479
SUE	0.001	0.065	−0.120	−0.014	0.002	0.013	0.113
Stocks held (per household)	2.263	2.406	1.000	1.000	1.000	4.000	12.00
Stocks traded (per month per household)	1.75	1.78	1.00	1.00	1.00	3.00	9.00
Number of papers that cover EA	0.879	1.534	0.000	0.000	0.000	3.000	7.00
Local Media Coverage (Dummy)	0.028	0.166	0.000	0.000	0.000	0.000	1.000

Read:

- Table I gives the descriptive backbone of the paper.
- Mean market capitalization is about \$5.66 billion; mean SUE is near zero; the typical household holds 2.263 stocks and trades 1.75 per month.
- The mean number of papers covering an earnings announcement is 0.879, but the mean of `Local Media Coverage` is only 0.028.
- The crucial descriptive message is the strong local tilt on both sides: investors hold local stocks disproportionately, and newspapers cover local firms disproportionately.
- This table tells you both why the design is promising and why endogeneity is dangerous.

5.2 Table II: the main local-media regressions

Table II

Media Effects and the Trading of Households

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities, which we call "trading areas." The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is available between January 1991 and November 1996 and is taken from the discount brokerage database in Barber and Odean (2000). Trading volume is considered on day 0, 1, or 2 following the earnings announcement date as identified by COMPUSTAT and I/B/E/S. *Local Media Coverage* takes the value one if the local newspaper (i.e., the newspaper that corresponds to the trading area) wrote a story about the firm on day +0, +1 or +2 following the earnings announcement. Controls include the natural logarithm of market capitalization (size), dummy variables for varying quintiles of earnings surprise (SUE quintiles), dummy variables for coverage in each of our local and national newspapers, dummy variables for each of the 19 trading areas and date fixed effects. Robust standard errors clustered by firm are in parentheses. *, **, and *** represent significance at the 10%, 5% and 1% levels, respectively.

	Dependent Variable: Log Dollar Trading Volume				
Local Media Coverage	0.746***	0.658***	0.648***	0.370***	0.477***
	(0.0830)	(0.0739)	(0.0739)	(0.0458)	(0.0593)
Firm Size		0.0700***	0.0761***	0.0506***	0.0577***
		(0.0103)	(0.0108)	(0.0100)	(0.0113)
SUE Quintile = Lowest			0.0758***	0.0381***	0.0526***
			(0.0158)	(0.0153)	(0.0171)
SUE Quintile = 2			0.0128	0.00223	0.00988
			(0.00950)	(0.0084)	(0.00921)
SUE Quintile = 4			0.0308**	0.00688	0.0235**
			(0.0119)	(0.0093)	(0.0107)
SUE Quintile = Highest			0.0845***	0.0502***	0.0661***
			(0.0185)	(0.0177)	(0.0197)
Coverage in *Boston Globe*				−0.235**	0.0305
				(0.101)	(0.0426)
Coverage in *Denver Post*				0.122	−0.00496
				(0.111)	(0.0501)
Coverage in *Detroit News*				−0.102	0.0814
				(0.0953)	(0.0641)
Coverage in *Houston Chronicle*				0.148	0.0257
				(0.0990)	(0.0431)
Coverage in *Las Vegas Review Journal*				−0.122	−0.0671
				(0.111)	(0.0474)
Coverage in *Los Angeles Times*				0.0416	0.0618*
				(0.0974)	(0.0356)
Coverage in *New York Times*				−0.0618	0.0341**
				(0.0386)	(0.0154)
Coverage in *Pittsburgh Post Gazette*				−0.00631	0.0891*
				(0.0860)	(0.0457)
Coverage in *San Antonio Express News*				0.00900	−0.0298
				(0.120)	(0.0419)
Coverage in *San Diego Union Tribune*				0.0384	0.0531
				(0.159)	(0.0612)
Coverage in *San Francisco Chronicle*				−0.420**	0.261***
				(0.208)	(0.0759)

(*continued*)

Table II—*Continued*

	Dependent Variable: Log Dollar Trading Volume				
Coverage in *Seattle Post Intelligencer*				−0.107	0.168
				(0.376)	(0.187)
Coverage in *St. Louis Post Dispatch*				0.00945	−0.0794**
				(0.0908)	(0.0381)
Coverage in *St. Petersburg Times*				−2.118***	0.0519
				(0.321)	(0.140)
Coverage in *Minneapolis Star Tribune*				0.0453	0.0580
				(0.0788)	(0.0401)
Coverage in *Atlanta Journal Constitution*				−0.188**	−0.0780**
				(0.0776)	(0.0353)
Coverage in *Sacramento Bee*				−0.166	−0.152
				(0.328)	(0.259)
Coverage in *Washington Post*				0.209**	0.0587**
				(0.0813)	(0.0288)
Coverage in *New Orleans Times-Picayune*				−0.251***	−0.104***
				(0.0850)	(0.0326)
Coverage in *USA Today*				0.0742	0.173***
				(0.120)	(0.0382)
Coverage in *Wall Street Journal*				−0.0733**	−0.00561
				(0.0347)	(0.0145)
Industry fixed effects	NO	NO	YES	YES	YES
City fixed effects	NO	NO	NO	NO	YES
Date fixed effects	YES	YES	YES	YES	YES
Observations	276,982	276,982	273,999	265,928	273,999
Adjusted R^2	0.026	0.039	0.041	0.040	0.058

Read:

- Table II is the baseline result.
- The coefficient on `Local Media Coverage` starts at 0.746, then becomes 0.658, 0.648, 0.370, and finally 0.477 as the authors add firm size, industry controls, paper controls, and city controls.
- The final specification is the main one to remember: local coverage is associated with about a 48% increase in local trading volume.
- The SUE coefficients are much smaller. The paper's explicit point is that, for retail investors, media coverage appears far more behaviorally powerful than the underlying surprise variable.
- Another useful detail: once paper controls are included, the few nonlocal newspapers with spillovers are the *New York Times* and the *San Francisco Chronicle*, which are plausibly read beyond their home cities.

5.3 Table III: buys versus sells

Read:

- Table III splits trades into buys and sells.

- In the fully controlled specification, the local-media coefficient is 0.290 for buys and 0.260 for sells.
- So the media effect is somewhat stronger for buys, but the sell-side response is also clearly there.
- This is important because it rejects a narrow reading in which media simply induces net positive demand.

Table III

Media Effects among Buys and Sells

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call "trading areas." The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is available between January 1991 and November 1996 and is taken from the discount brokerage database in Barber and Odean (2000). Trading volume is considered on day 0, 1, or 2 following the earnings announcement date as identified by COMPUSTAT and I/B/E/S. *Local Media Coverage* takes the value one if the local newspaper (i.e., the newspaper that corresponds to the trading area) wrote a story about the firm on day 0, 1, or 2 following the earnings announcement. The first five columns consider only buy volume while the second five columns only consider sell volume. Controls include the natural logarithm of market capitalization (size), dummy variables for varying quintiles of earnings surprise (SUE quintiles), dummy variables for coverage in each of our local and national newspapers, dummy variables for each of the 19 trading areas and date fixed effects. Robust standard errors clustered by firm are in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: Log Dollar Trading Volume (Only Buys)					Dependent Variable: Log Dollar Trading Volume (Only Sells)				
Local Media Coverage	0.462***	0.408***	0.402***	0.339***	0.290***	0.381***	0.341***	0.335***	0.307***	0.260***
	(0.0588)	(0.0524)	(0.0521)	(0.0428)	(0.0406)	(0.0464)	(0.0426)	(0.0430)	(0.0391)	(0.0374)
Firm size		0.0446***	0.0480***	0.0341***	0.0341***		0.0312***	0.0342***	0.0268***	0.0268***
		(0.00684)	(0.00714)	(0.00686)	(0.00686)		(0.00515)	(0.00540)	(0.00581)	(0.00581)
SUE Quintile = Lowest			0.0388***	0.0191*	0.0191*			0.0438***	0.0321***	0.0321***
			(0.0104)	(0.0107)	(0.0107)			(0.00864)	(0.00932)	(0.00932)
SUE Quintile = 2			0.000173	−0.00407	−0.00407			0.0150***	0.0130**	0.0130**
			(0.00716)	(0.00672)	(0.00672)			(0.00564)	(0.00561)	(0.00561)
SUE Quintile = 4			0.00898	0.00340	0.00340			0.0261***	0.0211***	0.0211***
			(0.00885)	(0.00786)	(0.00786)			(0.00690)	(0.00623)	(0.00623)
SUE Quintile = Highest			0.0457***	0.0309***	0.0309***			0.0469***	0.0386***	0.0386***
			(0.0120)	(0.0116)	(0.0116)			(0.0113)	(0.0123)	(0.0123)
Industry fixed effects	NO	NO	YES	YES	YES	NO	NO	YES	YES	YES
Paper fixed effects	NO	NO	NO	YES	YES	NO	NO	NO	YES	YES
City fixed effects	NO	NO	NO	NO	YES	NO	NO	NO	NO	YES
Date fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	276,982	276,982	273,999	273,999	273,999	276,982	276,982	273,999	273,999	273,999
Adjusted R^2	0.022	0.031	0.032	0.041	0.048	0.015	0.021	0.022	0.026	0.034

Table IV

Media Effects and Determinants of Local Trading

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call "trading areas." The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is taken from the discount brokerage database in Barber and Odean (2000). Trading volume is considered on day 0, 1, or 2 following the earnings announcement date as identified by COMPUSTAT and I/B/E/S. *Local Media Coverage* takes the value one if the local newspaper (i.e., the newspaper that corresponds to the trading area) wrote a story about the firm on day 0, 1, or 2 following the earnings announcement. *Local Firm Dummy* takes the value one if the firm is local to the trading market. *Fraction of Accounts Holding* is the fraction of accounts in the trading area holding the stock as of the first day of the month. *Fraction of Accounts Trading* is the fraction of accounts in the trading area that traded the stock in the prior month. Controls include the natural logarithm of market capitalization (size), dummy variables for varying quintiles of earnings surprise (SUE quintiles), dummy variables for coverage in each of our local and national newspapers, dummy variables for each of the 19 trading areas and date fixed effects. Robust standard errors clustered by firm are in parentheses. ** and *** represent significance at the 5% and 1% levels, respectively.

	Dependent Variable: Log Dollar Trading Volume				
	Only Nonlocal Firms				
Local Media Coverage	0.283***	0.327***	0.382***	0.454***	0.281***
	(0.0412)	(0.0527)	(0.0570)	(0.0569)	(0.0510)
Firm Size	0.0508***	0.0565***	0.0234***	0.0476***	0.0230***
	(0.00999)	(0.0112)	(0.00532)	(0.00922)	(0.00495)
SUE Quintile = Lowest	0.0384**	0.0479***	0.0283***	0.0401***	0.0255***
	(0.0154)	(0.0170)	(0.0101)	(0.0141)	(0.00940)
SUE Quintile = 2	0.00275	0.00740	0.00113	0.00532	0.000547
	(0.00838)	(0.00907)	(0.00789)	(0.00853)	(0.00772)
SUE Quintile = 4	0.00732	0.0212**	0.0231***	0.0197**	0.0212**
	(0.00934)	(0.0105)	(0.00873)	(0.00955)	(0.00839)
SUE Quintile = Highest	0.0509***	0.0622***	0.0474***	0.0568***	0.0449***
	(0.0178)	(0.0196)	(0.0124)	(0.0173)	(0.0119)
Local Firm Dummy		0.492***			0.336***
		(0.0598)			(0.0518)
Fraction of Accounts Holding			28.34***		23.16***
			(3.030)		(2.764)
Fraction of Accounts Trading				69.78***	50.79***
				(6.974)	(4.976)
Industry fixed effects	YES	YES	YES	YES	YES
Paper fixed effects	YES	YES	YES	YES	YES
City fixed effects	YES	YES	YES	YES	YES
Date fixed effects	YES	YES	YES	YES	YES
Observations	265,928	273,999	273,999	273,999	273,999
Adjusted R^2	0.049	0.068	0.081	0.073	0.089

5.4 Table IV: controlling directly for local investor demand

Read:

- Column 1 drops all local firms and still finds a local-media coefficient of 0.283. That already says the effect is not just a home-bias-to-local-headquarters result.
- Column 2 adds a local-firm dummy and finds a local-firm effect of about 0.492. Home bias is real, but the media effect survives.
- Column 3 adds the fraction of local accounts holding the stock; the coverage coefficient is 0.382 and the holdings control is huge at 28.34.
- Column 4 adds the fraction of local accounts trading the stock in the previous month; the coverage coefficient is 0.454 and the lagged-trading control is 69.78.
- In the final column, all three local-demand controls enter together. The media coefficient is still 0.281, with the local-firm dummy at 0.336, fraction holding at 23.16, and fraction trading at 50.79.

- This is the most direct rebuttal to the “papers only write what locals already care about” story.

5.5 Table V: weather and transmission

Read:

- Table V is one of the most persuasive causal tables in the paper.
- The baseline local-media coefficient remains about 0.478.
- But the interaction of local coverage with hail is −0.922, with snow is −0.774, and with the aggregate extreme-weather dummy is −0.790.
- The level effects of weather itself are weak. The interaction is the entire point: on days when physical delivery is disrupted, investors do not respond to the local story.
- That is very hard to reconcile with a pure omitted-variable explanation.

Table V

Media Effects, Newspaper Delivery, and Local Weather

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call “trading areas.” The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement and is taken from the discount brokerage database in Barber and Odean (2000). *Local Media Coverage* takes the value one if the local newspaper (i.e., the newspaper that corresponds to the trading area) wrote a story about the firm on day 0, 1, or 2 following the earnings announcement date as identified by COMPUSTAT and I/B/E/S. Weather data are taken from the National Climatic Data Center at ftp://ftp.ncdc.noaa.gov/pub/data/gsod. For each trading area, we use weather from the station closest to the local newspaper's zip code. *Local Hail* is a dummy variable that takes the value one if there was hail during the 2 days after the earnings announcement. *Local Snow* is a dummy variable that takes the value one if there was at least 8 inches of new snowfall during the 2 days after the earnings announcement. *Extreme Weather* is a dummy variable takes the value of one if either *Local Hail* or *Local Snow* take the value of one. Controls include the natural logarithm of market capitalization (size), dummy variables for varying quintiles of earnings surprise (SUE quintiles), dummy variables for coverage in each of our local and national newspapers, dummy variables for each of the 19 trading areas and date fixed effects. Robust standard errors clustered by firm are in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: Log Dollar Trading Volume		
Local Media Coverage	0.478***	0.478***	0.479***
	(0.0594)	(0.0593)	(0.0593)
Firm size	0.0459***	0.0565***	0.0565***
	(0.0104)	(0.0113)	(0.0113)
SUE quintile = Lowest	0.0381**	0.0484***	0.0484***
	(0.0157)	(0.0170)	(0.0170)
SUE quintile = 2	−0.00504	0.00754	0.00752
	(0.00913)	(0.00908)	(0.00908)
SUE quintile = 4	0.0193*	0.0216**	0.0216**
	(0.0114)	(0.0105)	(0.0105)
SUE quintile = Highest	0.0522***	0.0633***	0.0632***
	(0.0186)	(0.0197)	(0.0197)
Local Hail	0.0998		
	(0.0745)		
Local Hail * Local Media Coverage	−0.922***		
	(0.129)		
Local Snow		0.0486	
		(0.0352)	
Local Snow * Local Media Coverage		−0.774***	
		(0.135)	
Local Extreme Weather			0.0641*
			(0.0344)
Local Extreme Weather * Local Media Coverage			−0.790***
			(0.105)
Industry fixed effects	YES	YES	YES
Paper fixed effects	YES	YES	YES
City fixed effects	YES	YES	YES
Date fixed effects	YES	YES	YES
Observations	273,999	273,999	273,999
Adjusted R^2	0.051	0.063	0.063

Table VI

Media Effects and Alternative Definitions of Household Trading

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call “trading areas.” The dependent variable is some measure of trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is available between January 1991 and November 1996 and is taken from the discount brokerage database in Barber and Odean (2000). Trading volume is considered on day 0, 1, or 2 following the earnings announcement date as identified by COMPUSTAT and I/B/E/S. *Local Media Coverage* takes the value one if the local newspaper (i.e., the newspaper that corresponds to the trading area) wrote a story about the firm on day 0, 1 or 2 following the earnings announcement. In the first three columns, the dependent variable is a dummy variable that takes the value one if there was any trading in the firm making the earnings announcement. In the second three columns, the dependent variable is the natural logarithm of one plus the number of accounts that traded in the firm making the earnings announcement. Controls include the natural logarithm of market capitalization (size), dummy variables for varying quintiles of earnings surprise (SUE quintiles), dummy variables for coverage in each of our local and national newspapers, dummy variables for each of the 19 trading areas and date fixed effects. Robust standard errors clustered by firm are in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: Log Number of Accounts Traded			Dependent Variable: Local Trading Dummy		
	All Trades	Buys Only	Sells Only	All Trades	Buys Only	Sells Only
Local Media Coverage	0.0467***	0.0272***	0.0230***	0.0501***	0.0311***	0.0277***
	(0.00673)	(0.00427)	(0.00369)	(0.00591)	(0.00421)	(0.00389)
Firm size	0.00487***	0.00281***	0.00224***	0.00604***	0.00368***	0.00286***
	(0.00103)	(0.000576)	(0.000531)	(0.00118)	(0.000729)	(0.000613)
SUE quintile = Lowest	0.00422***	0.00153*	0.00279***	0.00556***	0.00226*	0.00363***
	(0.00156)	(0.000926)	(0.000832)	(0.00182)	(0.00116)	(0.00101)
SUE quintile = 2	0.000766	−0.000240	0.00107**	0.000937	−0.000387	0.00148**
	(0.000800)	(0.000588)	(0.000457)	(0.000993)	(0.000738)	(0.000617)
SUE quintile = 4	0.00203**	0.000382	0.00179***	0.00220**	0.000269	0.00224***
	(0.000965)	(0.000735)	(0.000512)	(0.00110)	(0.000833)	(0.000660)
SUE quintile = Highest	0.00544***	0.00241**	0.00327***	0.00669***	0.00327***	0.00411***
	(0.00183)	(0.00102)	(0.00111)	(0.00205)	(0.00123)	(0.00130)
Industry fixed effects	YES	YES	YES	YES	YES	YES
Paper fixed effects	YES	YES	YES	YES	YES	YES
City fixed effects	YES	YES	YES	YES	YES	YES
Date fixed effects	YES	YES	YES	YES	YES	YES
Observations	273,999	273,999	273,999	273,999	273,999	273,999
Adjusted R^2	0.070	0.054	0.035	0.060	0.047	0.033

5.6 Table VI: alternative definitions of trading

Read:

- Table VI checks whether the results are being driven by a few large trades.

- When the dependent variable is *log number of accounts traded*, local-media coefficients are 0.0467 for all trades, 0.0272 for buys, and 0.0230 for sells.
- When the dependent variable is a *dummy for any local trade*, coefficients are 0.0501 for all trades, 0.0311 for buys, and 0.0277 for sells.
- So local stories bring in more traders and raise the probability of any trading at all. The effect is not just order-size inflation.

5.7 Table VII: pairwise fixed effects

Read:

- Table VII is the "sweep out everything possible" table.
- With only firm–date fixed effects, the coefficient is 0.477.
- With only firm–city fixed effects, it falls sharply to 0.0755. This shows that persistent local preferences matter a lot.
- With only city–date fixed effects, it is again large at 0.473.
- With all three pairwise fixed effects together, the coefficient is still 0.0796.
- The right interpretation is not that the media effect disappears; it is that after controlling for an enormous amount of latent heterogeneity, local coverage still predicts about 8% more local trading.

Table VII

Media Effects and Robustness

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call "trading areas." The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is available between January 1991 and November 1996 and is taken from the discount brokerage database in Barber and Odean (2000). Controls include Firm-Date, Firm-City and City-Date fixed effects. Robust standard errors clustered by firm are in parentheses. ** and *** represent significance at the 5% and 1% levels, respectively.

	Dependent Variable: Log Dollar Trading Volume			
Local coverage	0.477***	0.0755**	0.473***	0.0796***
	(0.0587)	(0.0295)	(0.0535)	(0.0281)
Firm-date fixed effects	YES	NO	NO	YES
Firm-city fixed effects	NO	YES	NO	YES
City-date fixed effects	NO	NO	YES	YES
Observations	273,999	273,999	273,999	273,999
Adjusted R^2	0.093	0.164	0.106	0.289

Table VIII

Media Effects, the Timing of Stories, and Falsification

Every firm earnings announcement in our sample corresponds to 19 distinct observations representing trading in each of 19 major U.S. cities that we call "trading areas." The dependent variable is the natural logarithm of one plus the dollar trading volume in each trading area for the firm that makes the earnings announcement where local trading volume is available between January 1991 and November 1996 and is taken from the discount brokerage database in Barber and Odean (2000). In column 1 (2, 3) only trading volume on day 0 (1, 2) following the earnings announcement is considered. Local Coverage on Day 0 (1, 2) takes the value one if there was a news article in the local newspaper on day 0 (1, 2). Local Coverage on Non-Matched Day takes the value one if there was a news article on a different day from the trading day (1 day after or 2 days after). Controls include Firm-Date, Firm-City, and City-Date fixed effects. Robust standard errors clustered by firm are in parentheses. ** represents significance at the 5% levels.

	Dependent Variable: Log Dollar Trading Volume		
	On Day 0	On Day 1	On Day 2
Local coverage on Day 0	0.0915**	−0.0154	−0.00914
	(0.0355)	(0.0282)	(0.0240)
Local coverage on Day 1	0.0112	0.0688**	0.0111
	(0.0316)	(0.0287)	(0.0193)
Local coverage on Day 2	0.0161	0.00941	0.0643
	(0.0395)	(0.0551)	(0.0408)
Firm-date fixed effects	YES	YES	YES
Firm-city fixed effects	YES	YES	YES
City-date fixed effects	YES	YES	YES
Observations	273,999	273,999	273,999
Adjusted R^2	0.269	0.264	0.222

5.8 Table VIII: timing of stories and falsification

Read:

- Table VIII is the high-frequency falsification exercise.
- For trading on day 0, only day-0 coverage matters: 0.0915. Coverage on day 1 or day 2 does not predict day-0 trading.

- For trading on day 1, only day-1 coverage matters: 0.0688. Neither day-0 nor day-2 coverage matters.
- For trading on day 2, only day-2 coverage matters in economic terms: 0.0643, while earlier-day coverage is tiny.
- This is exactly the pattern a causal transmission story predicts. It is also exactly the pattern an omitted-demand story should have trouble matching.

6 Short summary

- The paper studies whether media coverage causally changes investor trading behavior.
- It compares retail investors in 19 local markets reacting to the same S&P 500 earnings announcements but facing different local-newspaper coverage.
- The baseline result is large: local newspaper coverage increases local trading by roughly 8% to 50%, depending on specification.
- The effect appears for both buys and sells, survives detailed controls for local demand, survives saturated fixed effects, disappears when weather disrupts newspaper delivery, and lines up exactly with the day the story is printed.
- The paper's core message is therefore causal: media affects financial-market behavior by shaping which information reaches which investors and when.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that local newspaper coverage has a causal effect on local retail trading around earnings announcements. The central evidence is not one coefficient but a whole structure of results: the effect survives controls, survives fixed effects, vanishes when delivery is disrupted, and appears on the exact day the news arrives.

7.2 Economic intuition

Information in financial markets is not just "out there." It reaches investors through transmission channels, and those channels matter. Local newspapers lower the cost of noticing, processing, and acting on a firm's earnings release. When the local paper covers the story, more local investors trade; when the paper does not cover it, the same underlying event produces a weaker local response.

7.3 Takeaway

The broad lesson is methodological as well as substantive. Methodologically, the paper shows how to identify media effects in finance by holding the event fixed and varying exposure. Substantively, it shows that the media is part of market structure itself: not just a reflector of information, but a causal determinant of who reacts, where, and when.